Steve Yeh

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Placement Directors

Professor Mark Dean — mark.dean@columbia.edu

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Graduate Administrators

Amy Devine — aed2152@columbia.edu

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Education

Ph.D. in Economics, Columbia University. 2021–2027 (expected)

Committee: Evan Sadler (advisor), Navin Kartik, Qingmin Liu.

M.Phil in Economics, Columbia University. 2021–2024

M.A. in Economics, Columbia University. 2021–2023

B.A. in Economics and Mathematics, Cornell University 2017–2021

Summa Cum Laude in Economics and *Magna Cum Laude* in Mathematics.

Minors in Computer Science and Law & Society.

Research Fields

Networks, Microeconomic Theory.

Job Market Paper

A Network Theory of Project Management

Abstract: Managers often have little discretion over pay, but they routinely decide who works on which projects. I study a manager who staffs projects by capping each worker's effort on each project. Workers prefer some projects to others and work harder when their teammates do, but find spreading effort across projects costly. Optimal assignments are characterized by a single ranking of projects, and two worker-specific cutoffs on it determine where each worker contributes freely, is held below his desired effort, or is kept off a project team. Because the ranking is common to all workers, project teams are nested. Standard network centrality can get the ordering of generalists and specialists backwards because the links that make a worker valuable to one team also make spreading him across several teams costly. Workers instead sort by a ratio of two centralities that captures their comparative advantage in spreading versus concentrating effort. Assignment selects the network through which shocks propagate, and an improvement at one project can degrade every other project in the organization.

Working Papers

Optimal Project Task Assignments

Revise & Resubmit at *Theoretical Economics*

Abstract: A principal assigns agents to tasks in a multi-task project. Agents tend to shirk at their tasks and rely on others' efforts because effort on one task substitutes for effort on similar tasks. Assigning the same agent to more tasks mitigates free-riding, but convex effort costs prevent him from completing all of his tasks. For a class of projects, an optimal task assignment assigns each agent to a module comprising a task central to the project and all locally related tasks. For general projects, modular task assignments are approximately optimal with a performance guarantee that depends on a project's task structure.

A Shocking Reversal: Flipping Complements and Substitutes in Network Games (with Evan Sadler)

Extended Abstract in the *Proceedings of the 27th ACM Conference on Economics and Computation*.

Abstract: In conventional network games, each player chooses one action that impacts all of her neighbors. We study players that select neighbor-specific efforts and face a convex total effort cost. Our main result highlights a striking reversal in comparative statics: When bilateral efforts are strategic complements (substitutes), shocks propagate through the network in a pattern befitting strategic substitutes (complements) in standard network games. This reversal, together with other fundamental changes in player behavior, suggests that retaining the single effort assumption when it is not appropriate may lead to poor policy recommendations.

Plausible Pairwise Stability

Abstract: I propose a stability concept for network formation under incomplete information. A network is plausibly pairwise stable if players' beliefs are consistent with common knowledge of network stability. I provide a general condition under which plausibly pairwise stable networks exhibit clustering. In a simple model of vertical differentiation, I characterize minimal plausibly pairwise stable networks. I show that incomplete information can stabilize segregated and sparse networks that are unstable under complete information. I discuss how incomplete information resolves the tension between stability and efficiency in the presence of negative linking externalities.

Referee Service

Program Committee for ACM Conference on Economics and Computation (2026), Journal of Economic Theory.

Presentations

2026: Workshop on Advances in Economic Theory and Beyond (Tsinghua), Economics of Networks Workshop (Stony Brook), 37th Stony Brook International Conference on Game Theory, 27th ACM Conference on Economics and Computation (Rome), North American Summer Meeting of the Econometric Society (Emory), 11th Annual Conference on Network Science and Economics (University of Miami).

2025: 36th Stony Brook International Conference on Game Theory.

2024: Northwestern-Kellogg Summer School in Economic Theory, 35th Stony Brook International Conference on Game Theory.

Teaching

Intermediate Microeconomics (Undergraduate) — Spring 2023, Fall 2026
TA for Prof. Ingmar Nyman (2023) and Prof. Caterina Musatti (2026).

Game Theory (Undergraduate) — Spring 2025
TA for Prof. Babak Somekh.

Microeconomic Analysis II (MA) — Spring 2024
TA for Prof. Guillaume Haeringer.

Principles of Economics (Undergraduate) — Fall 2023
TA for Prof. Waseem Noor.

Microeconomic Analysis I (MA) — Fall 2022, Fall 2024
TA for Prof. Bernard Salanié (2022) and Prof. Lilia Maliar (2024).

Microeconomics I (MPA) — Summer 2023, 2024, 2025, 2026
TA for Prof. Muhammad Asali.

Math Camp Instructor (MA, MPA, MIA) — Summer 2022, 2023, 2024, 2025

Awards

Dean's Fellowship, Columbia University — 2021–Present

Wueller Fellowship for Best Teaching Fellow Runner-Up, Columbia University — 2024

Harriss Award for Best 2nd-Year Paper Runner-Up, Columbia University — 2023

Uri M. Possen Memorial Award for Best Undergraduate Honors Thesis, Cornell University — 2021

Raymond & Bernice Conrad Endowed Scholarship, Cornell University — 2019–2021

New York State Scholarship for Academic Excellence, Cornell University — 2017–2021

Other Information

Employment: Goldman Sachs, Credit Risk Intern (Summer 2020).

Citizenship: US.

Languages: English (Native), Hakkanese (Native), Mandarin (Fluent), Spanish (Proficient).

References

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